



Partitioning Soil Respiration: Differentiating Root Contributions and Environmental Factors

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Abstract

An essential factor for understanding root respiration, in addition to carbon dioxide emission, is the soil oxygen influx, thus the objective was to evaluate carbon dioxide (CO₂) emissions and oxygen (O₂) influx (soil respiration) in soils with and without vegetation, understanding the contribution of root respiration to total soil respiration and its correlation with meteorological variables. Monitoring took place with readings at 7, 10, 12, 15 and 17 h (system 24 h) in different environmental conditions. To design the experiment, two plots of 1m² were adopted under the conditions of bare soil and soil vegetated with grass (*Paspalum Notatum*). To determine the emission of CO₂ from the soil, the Li-8100 portable system was used, while the influx of O₂ from the soil was obtained by linear interpolation of the oxygen variation in the chamber using a smart oxygen sensor UV Flux 25% model CM-42,951. The data were subjected to analysis of variance for repeated measures over time. The results demonstrated that CO₂ emission and O₂ influx are higher in vegetated soil. The peak CO₂ emission and the peak oxygen influx occur between 10 and 15 h. CO₂ emission was strongly associated with soil moisture without vegetation ($r=0.76$; $p<0.05$) and with vegetation ($r=0.80$, $p<0.05$). The soil O₂ influx in the vegetated soil was associated with photosynthetically active radiation ($r=0.85$, $p<0.05$) and is linked to the activity of photosynthesis and consequently root respiration. The existence of root respiration and its dependence on the soil O₂ influx are concluded, as the O₂ influx is increased in vegetated soils in relation to bare soil. And finally, root respiration represented 27 to 48% of soil respiration for the evaluated scenario.

Keywords Root respiration · Soil management · CO₂ emission · O₂ influx · Climate change

Introduction

Soil respiration is a fundamental environmental process that regulates the carbon cycle and drives the decomposition of soil organic matter. Over the past decades, numerous studies have explored the factors controlling soil respiration, including temperature, soil moisture, and soil properties [28, 82, 84]. More recently, attention has shifted toward understanding soil gas flux dynamics due to the increasing concentration of greenhouse gases (GHGs) in the atmosphere, particularly carbon dioxide (CO₂), which contributes to global warming [8, 17, 80].

Although microbial respiration represents a major component of total soil respiration, the long-term effects of increased carbon inputs on soil microbial biomass (SMB) remain uncertain [5]. Furthermore, the interactions between SMB and plant roots are often neglected [11, 12, 52]. Consequently, the role of root respiration and rhizosphere

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processes has been underestimated in studies of soil carbon dynamics [37, 56].

Another aspect that deserves attention is soil oxygen (O₂) influx, which supports cellular respiration and is critical for root metabolism [2, 10, 48]. Only a few studies have quantified O₂ influx and its relationship with CO₂ emissions, suggesting that root activity plays a major role in regulating both processes [21, 55, 72]. However, the contribution of O₂ exchange to improve soil respiration predictions remains poorly investigated [31, 62].

Root respiration is directly linked to photosynthetic activity, since carbon fixed by photosynthesis provides the substrate for root metabolism [5, 24]. Environmental conditions such as temperature, soil moisture, and photoinhibition modulate this process by enhancing or restricting cellular respiration [21, 37]. Despite this close coupling, there is still limited understanding of how root respiration contributes to total soil CO₂ fluxes under variable environmental conditions.

Long-term field measurements are essential to improve the prediction of soil respiration and its response to climate change [83]. These analyses must integrate microbial community dynamics, soil type, vegetation cover, and climatic variables [9, 27, 54]. However, few studies have continuously monitored the relative contributions of root and microbial respiration in situ, and available estimates remain highly variable from 10 to 90% [36], around 50% [32, 49], or 20 to 50% [2] reflecting the strong influence of vegetation type and root density.

Land use is another important driver of soil respiration dynamics. Changes such as deforestation, intensive tillage, or urbanization alter soil organic matter and root biomass, modifying CO₂ efflux patterns [22, 40, 48, 60, 62]. These transformations highlight the need to quantify how root-driven respiration responds under contrasting land uses.

In summary, understanding gas exchange between CO₂ release and O₂ consumption in distinct soil compartments is crucial to improve global CO₂ emission models and to assess the effects of climatic variability and land-use change on carbon dynamics [39, 45, 63].

Despite these advances, this study has some inherent limitations related to its local scale and short-term monitoring period, which restrict the extrapolation of results. Nevertheless, the experimental design provides valuable insights into the relative contribution of root respiration to total soil CO₂ fluxes under field conditions.

We hypothesize that root respiration is closely associated with photosynthetic activity and exhibits diurnal variability linked to environmental factors. To test this, we compared CO₂ emissions and O₂ influx between bare soil minimizing rhizosphere influence and vegetated soil dominated by grasses with a dense root system. The objective was to

evaluate the contribution of root respiration to total soil respiration and its relationship with meteorological variables.

Materials and Methods

Location and Characterization of the Experimental Area

The experiment was carried out at the agroclimatological station of the Faculty of Agricultural and Veterinary Sciences (FCAV), campus of the São Paulo State University (UNESP), located in the municipality of Jaboticabal, west of the state of São Paulo, Brazil (Fig. 1). The station is located under the geographic coordinates 21°14'05" S and 48°17'09" W, with an altitude of 615 m.

The regional climate was classified as B1rB'4a', humid mesothermal, with small water deficit [70, 71]. The soil of the experimental area was classified by Brazilian System of Soil Classification (SiBCS) as Eutrophic Red Latosol Typical with clayey texture [15, 66]. Equivalence to Clayey Oxisol by Soil Taxonomy [6]. The physicochemical attributes of the soil for the surface layer (0.00 to 0.20 m) were presented by [23].

Experimental Design

The experiment was allocated in an area of 4.0 m² and was divided into four plots of 1.0 m², two plots of bare soil and two other plots of soil vegetated with potato grass (*Paspalum Notatum*) in each of the Four PVC collars measuring 0.05 m in height and 0.10 m in diameter were allocated to the plots. The treatment called bare soil was vegetated and the vegetation was removed to set up the experiment and PVC collars were installed on the soil 72 h before the first reading moment (Fig. 1). During the evaluation period, manual control was carried out in the so-called bare soil plots, thus maintaining the initial state of the treatment.

The readings were recorded at three different times with contrasting weather conditions and frequently occurring throughout the year. The first moment the reading occurred after a rain of 19.9 mm (02/10/2023) equivalent to day of the year number 275. The second moment after a period of drought, the readings occurred on 24/10/2023, equivalent to the day of the year number 297. The third moment is 11/09/2023 equivalent to the day of the year number 313, with the soil CO₂ emission, soil O₂ influx, soil moisture and soil temperature for bare soil (BS) and vegetated soil (VS) at the following times: 7, 10, 12, 15 and 17 h (system 24 h).

Data such as air temperature (Temp), relative humidity (RH), atmospheric pressure (AtmP) and photosynthetically active radiation (PAR) were also monitored at the time of

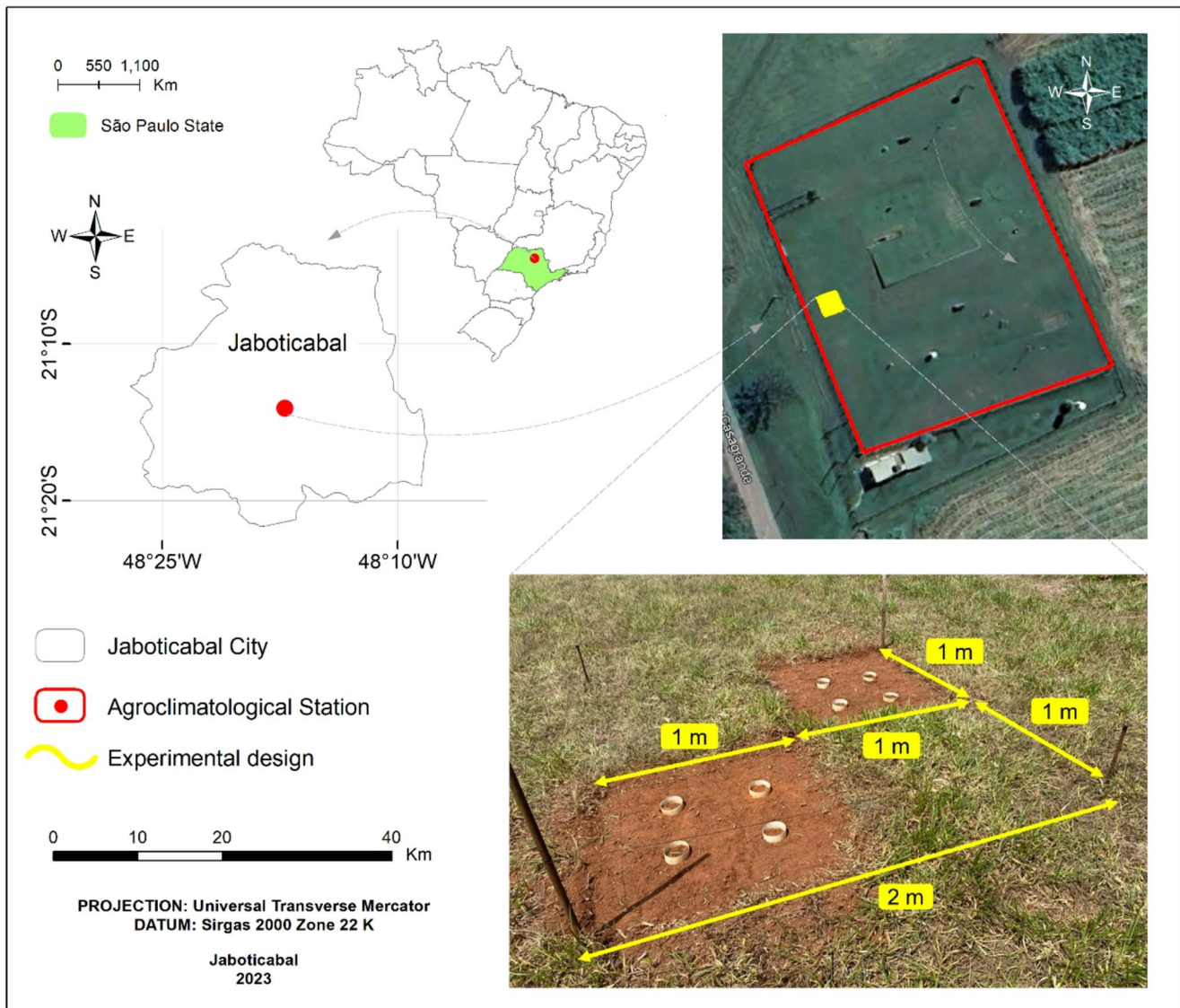


Fig. 1 Geographic information and experimental design

readings and were extracted from a set of data belonging to the collection of the Department of Engineering and Exact Sciences. The observations carried out at the Agroclimatological Station on the Jaboticabal Campus were collected, typed into a standardized format, and consistency and quality control were carried out. The variability of these meteorological variables for each moment in each of the times were presented in Table S1 (Supplementary).

Evaluation of Soil CO₂ Emission, Soil Moisture and Temperature

Soil CO₂ emission (FCO₂) readings were recorded using a portable LI-8100 system called Automated Soil CO₂ Flux System manufactured by LI-COR Biosciences®, Nebraska, USA. The LI-8100 system was coupled to a model

8100–102 camera that provides measurements on soil CO₂ emission. This chamber has an internal volume of 854.2 cm³ with a circular contact area of 83.7 cm². This system made it possible to verify changes in CO₂ concentrations, using the IRGA sensor (Infrared Gas Analyzer), through optical absorption spectroscopy in the near infrared spectral region. The chamber was placed on PVC collars previously inserted into the soil (72 h before) at each sampling point at a depth of 0.03 m. To determine soil CO₂ emission, the reading standard adopted was 90 s of reading with one-second reading intervals. The result of the variation of carbon dioxide in relation to time inside the chamber was obtained in μmol m⁻² s⁻¹.

Soil moisture was determined using TDR (Time Domain Reflectometry) equipment, a portable soil moisture sensor model HidroSense II (HS2) manufactured by Campbell

Scientific[®] that evaluated available soil moisture in percentage (volume/volume) using a double 0.12 m rod. Soil temperature was obtained using a rod thermometer inserted into the soil at a depth of 0.10 m next to the previously installed PVC collars. Soil temperature was obtained and presented in degrees Celsius.

Determination of Soil O₂ Influx

The determination of the soil of O₂ influx (iO₂) was carried out using a PVC camera coupled to a Smart Oxygen Sensor UV Flux 25% model CM-42,951 manufactured by CO₂ METER[®]. This system consists of a portable sensor that measures the variation in oxygen inside the camera using the principle of fluorescence with ultraviolet light. This system provides low bias and cross sensitivity. Thus, the camera with the sensor was coupled to a portable computer by the USB port and connected using the GasLab[®] software to record data in real time.

The soil O₂ influx was calculated by linear interpolation of the oxygen concentration values inside the chamber during 300 s (5 min) of sampling and considering the atmospheric pressure, the absolute air temperature in Kelvin and the camera volume. Obtaining the decay of O₂ inside the camera during the sampled period. The influx of O₂ from the soil was obtained by Eq. 1 and was presented in mg O₂ m⁻² s⁻¹.

$$iO_2 = \frac{dO_2}{dt} \frac{PM}{RT} H \quad (1)$$

where, dO_2/dt is the amount of O₂ (ppm) measured as a function of time t (s); P is atmospheric pressure (Pa); M is the molar mass of O₂ (g mol⁻³); R is the universal gas constant (8.314462 J mol⁻¹ K⁻¹); T is the absolute temperature (K) and $H = V/A$, for volume (V) = 6.6×10^{-6} m³ and collar area (A) = 0.008 m² above the ground [1, 22, 74].

Data Analysis

Analysis of Temporal Variability Initially, the basic assumptions were verified and met, thus, the residuals of the variables are normally distributed and were tested based on the Shapiro-Wilk normality test ($p > 0.05$) [69] and the hypothesis of homoscedasticity (equality between variances), was also tested based on the Bartlett test ($p > 0.05$) [7].

Once the requirements were met, data on soil CO₂ emission (FCO₂), soil oxygen influx (iO₂), soil temperature (ST) and soil moisture (SM) were subjected to analysis of diurnal temporal variability, which was evaluated using the analysis of variance method for experiments with repeated

measurements over time, this means that the different treatments (bare and vegetated soil) were evaluated as a function of time (in hours) [22, 50, 73], therefore, the averages were calculated for each assessment time. For multiple comparisons between means, the Tukey's Test.

Dynamic Relationship of Gases with the Environment

The relationship between the dynamics of gas flows in the soil and environmental conditions was analyzed based on two analyses. Initially, a linear correlation analysis was carried out between the gas flows in the soil (emission of carbon dioxide, influx of oxygen) with the meteorological variables collected at the agroclimatological station (air temperature, relative humidity, atmospheric pressure, photosynthetically active radiation), in addition to soil temperature and humidity and were performed using Pearson's linear correlation method and a significance level of 5% probability was adopted [702].

Hierarchical partition analysis (HPA) [16, 22] was then applied, which partitions the independent and joint contributions of each predictor in a set of multivariate data for a linear model by hierarchical decomposition. Meanwhile, the randomization test (at 5% probability) allows the evaluation of the most important scores for each assessment in explaining the response variable [87]. This ranking used by HPA analysis allows the selection of variables with greater security, as it focuses on the explanatory capacity of the phenomenon investigated.

Simple Linear Regression with Cross Validation K-fold of the Repetition

Regression analysis is a way of predicting the result of the study variable based on another predictor variable. To carry out this analysis, the results of the hierarchical partition analysis (mentioned previously) were considered. The simple linear regression model (LRM) is given by Eq. 2 [87].

$$y = \beta_1 X_0 + \beta_0 + \epsilon \quad (2)$$

where, y is the dependent variable of the linear regression equation; β_1 is the slope of the linear regression line; X_0 is the independent variable of the linear regression model; β_0 is the y-intercept of the linear regression equation; ϵ is the linear regression error.

The K-fold cross-validation method with repetition was used, which is a technique that allows the training and testing data to be divided into K times and then a resampling

of the data, called repetition, is applied. Thus, the number of folds equal to three was adopted (that is, the database was divided into three equal parts) with one hundred repetitions (it was resampled 100 times with the seed fixed), the data division parameter adopted was two training thirds and testing thirds in each resampling. The final regression model was obtained together with the average adjusted R^2 (accuracy measure) of all validated models, in addition to the mean squared error – RMSE (Root Mean Square Error) as an error measure. The use of this approach was justified because it is a more robust method in controlling trend and bias errors (Stone, 1978). All statistical analyzes were performed in R language [61]. With the help of the packages: *ExpDes.pt* [30], *factoextra* [43, 44], *readxl* [80], *tidyverse* [79], *patchwork* [57], *nortest* [34], *lawstat* [33], *caret* [46], *corrplot* [76] and *ggplot2* [76, 78].

Results

Diurnal Temporal Variability and Contribution of Root Respiration

For the first moment (DOY 275), at the different times evaluated, the emission of CO_2 from the soil was higher in the vegetated soil (Fig. 2a). Although the bare soil recorded the lowest soil CO_2 emission values when compared to the

vegetated soil, the emission at 10, 12 and 15 h (system 24 h) did not differ from each other ($7.56, 7.67$ and $7.16 \mu\text{mol m}^{-2} \text{s}^{-1}$, respectively), but they are higher than the emissions at 7 and 17 h, these did not differ from each other (5.01 and $5.26 \mu\text{mol m}^{-2} \text{s}^{-1}$, respectively).

Soil CO_2 emission maintained the pattern observed in the first moment and registered significant differences between vegetated soil and bare soil (BS) at all times evaluated (Fig. 2b). Between 7 and 17 h the vegetated soil recorded the following average values: $3.56, 4.70, 4.99, 4.29$ and $3.99 \mu\text{mol m}^{-2} \text{s}^{-1}$, respectively. However, the highest values occurred at 10 and 12 h and differed from other times. While bare soil recorded the following values: $2.46, 2.89, 3.33, 3.25$ and $3.04 \mu\text{mol m}^{-2} \text{s}^{-1}$, respectively for the same time interval. Although the bare soil had its emission peak at 12 o'clock, the values from 10 to 17 h did not differ from each other, but present significant differences for the emission at 7 am on the bare soil.

And finally, in the third moment (DOY 313), the emission of CO_2 from the soil followed the same pattern as observed in the previous moments (DOY 275 and 297), that is, it remained higher and more significant in the vegetated soil ($6.36, 8.34, 8.35, 7.90$ and $7.93 \mu\text{mol m}^{-2} \text{s}^{-1}$ respectively), in relation to bare soil ($3.24, 4.18, 4.36, 4.00$ and $4.22 \mu\text{mol m}^{-2} \text{s}^{-1}$ respectively) at all times investigated (Fig. 2c). It is worth noting that in the treatments, soil CO_2 emission

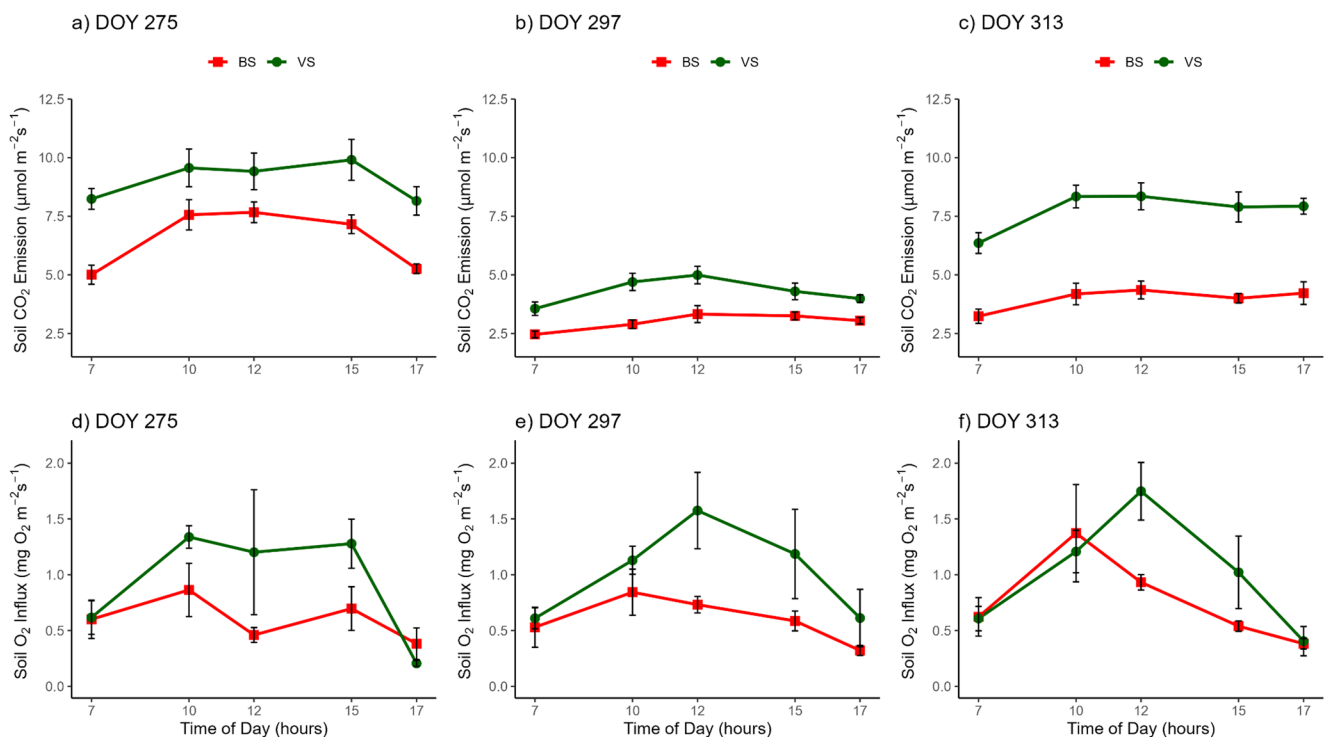


Fig. 2 Temporal analysis for soil CO_2 emission (in a–c) ($\mu\text{mol m}^{-2} \text{s}^{-1}$) and soil O_2 influx (in d–f) ($\text{mg O}_2 \text{ m}^{-2} \text{s}^{-1}$) for different day of year (DOY). The error bars represent the standard error of the mean. Note: Time system 24 h. BS is bare soil, VS is vegetated soil

increased from 10 am in relation to the values observed at 7 am (Fig. 2a).

In relation to the soil O_2 influx, it varied between treatments at the times evaluated (Fig. 2d). Although there are differences on the O_2 influx in vegetated soil (VS) and in bare soil (BS), the highest values in both cases occurred between 10 and 15 h, but there are no significant differences on the soil O_2 influx of between the treatments at the times of 7 and 17 h. The vegetated soil between 10 and 15 h recorded the following values for the soil O_2 influx: 1.34, 1.20 and 1.28 $mg\ O_2\ m^{-2}\ s^{-1}$, respectively. However, these values differed from the bare soil which recorded 0.86, 0.46 and 0.70 $mg\ O_2\ m^{-2}\ s^{-1}$, respectively between 10 and 15 h.

In the second moment (DOY 297), for the soil O_2 influx, not significant differences were observed between the treatments at the following times: 7, 10 and 17 h (Fig. 2e). On the other hand, at 12 and 15 h the soil O_2 influx was greater in the vegetated soil (1.58 and 1.19 $mg\ O_2\ m^{-2}\ s^{-1}$) compared to the bare soil (0.73 and 0.59 $mg\ O_2\ m^{-2}\ s^{-1}$) respectively for the same time. Following the same pattern as in the first moment (DOY 275), in the second it was also observed that the lowest values of soil O_2 influx for both treatments occurred at 7 and 17 h.

At the third moment (DOY 313), the O_2 influx from the vegetated soil (1.75 and 1.02 $mg\ O_2\ m^{-2}\ s^{-1}$) was higher and significant than the bare soil (0.93 and 0.54 $mg\ O_2\ m^{-2}\ s^{-1}$) at 12 and 15 h (Fig. 2f). While 7, 10 and 17 h did not

differ between treatments, this behavior was also observed in the second moment (DOY 297). Between 7 and 17 h, the highest values of soil O_2 influx occurred between 10 am and 15 h for the treatments investigated (Fig. 2f).

Soil moisture (SM) varied over time (DOY 275) (Fig. 3a). Soil moisture at 7 am did not differ between treatments (15.8 and 15.3% BS and VS, respectively), this behavior was also observed at 17 h (11.3 and 11.7% BS and VS, respectively). It was observed that in both treatments soil moisture decreased when observing the values at the beginning and end of the evaluations ($p < 0.05$). The moisture in bare soil was higher than in vegetated soil between 10 and 15 h ($p < 0.05$). The moisture values on the bare soil between 10 and 15 h were: 15.1; 14.0 and 13.0% and differed from the vegetated soil, and were; 13.6; 12.7 and 12.1%, respectively.

Soil moisture in the second moment (DOY 297) showed a different behavior than the first, instead of decreasing throughout the day, it maintained a stable range with small fluctuations. No significant differences were observed between the treatments at 10, 15 and 17 h, however they differed at 7 and 12 h, and in the morning the bare soil recorded a higher soil moisture content (3.4%); however, this was reversed at midday (12 h) with vegetated soil (3.9%) being higher than bare soil (2.8%) (Fig. 3b).

Soil moisture for the third moment (DOY 313) varied between the investigated treatments (Fig. 4c), being higher in bare soil (11.5, 10.4, 9.5, 9.8 and 8.7%) when compared

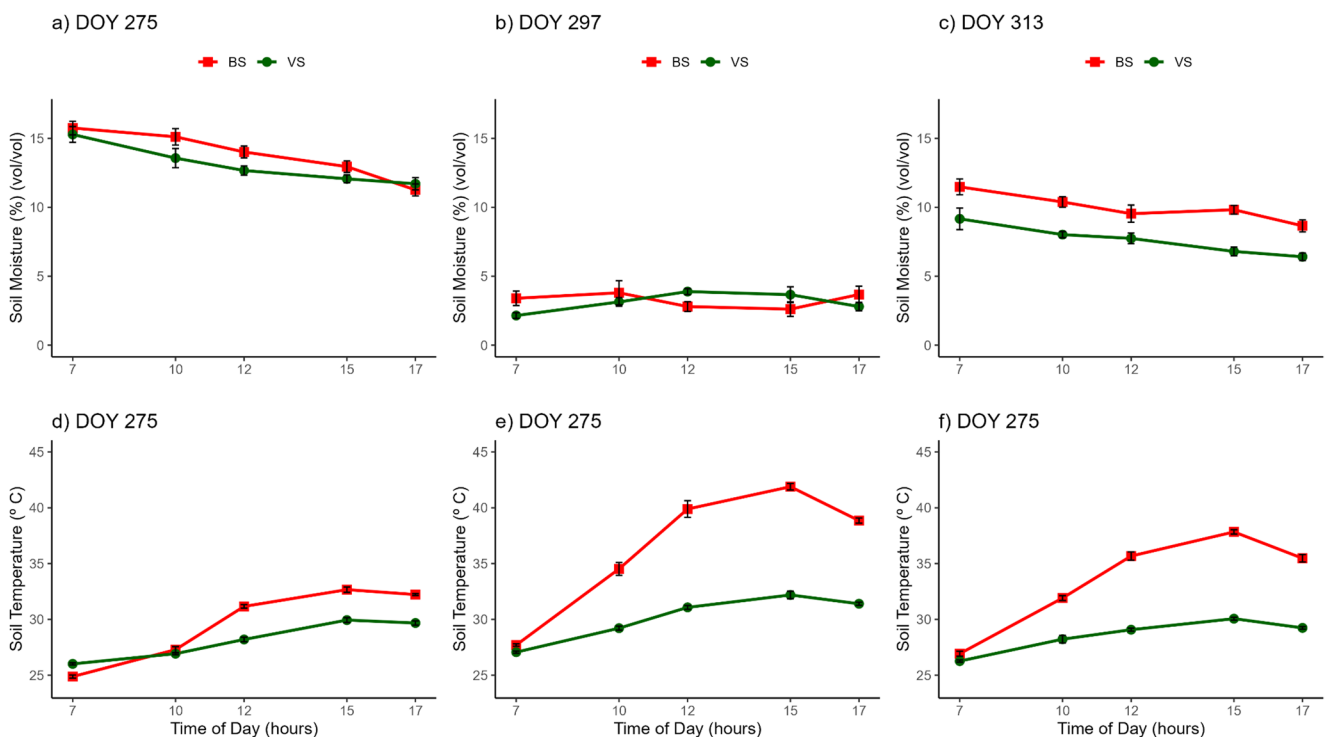


Fig. 3 Temporal analysis for soil moisture (in a–c) (%) (volume/volume) and soil temperature (in d–f) (°C) for different day of year (DOY). The error bars represent the standard error of the mean. Note: Time system 24 h. BS is bare soil, VS is vegetated soil

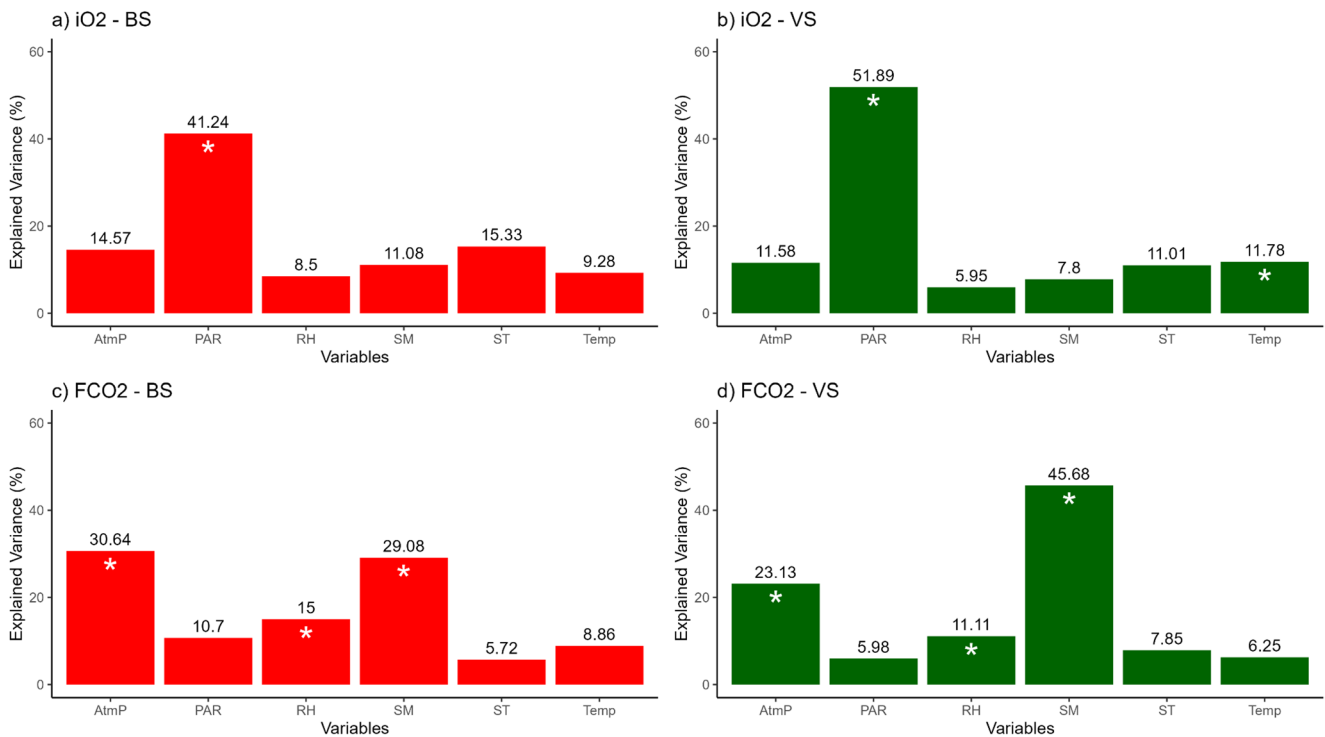


Fig. 4 Hierarchical partitioning analysis for soil O₂ influx (**a, b**) and soil CO₂ emission (**c, d**) under bare soil (BS) and vegetated soil (VS). *Asterisks indicate significance for the randomization test at 5% probability ($p < 0.05$)

to vegetated soil (9.2, 8.0, 7.7, 6.8 and 6.4%). However, soil moisture decreased throughout the day (Fig. 3c), so at 17 h (5 pm) soil moisture was lower compared to 7 am in both treatments, this decrease represented 24.4% for bare soil and 30.4% for bare soil. the vegetated soil.

Soil temperature (ST) varied throughout the day (Fig. 3d), but there were no significant differences at 10 am between the investigated treatments. At 7 am the temperature of the vegetated soil (26.0 °C) was higher than that of the bare soil (24.9 °C). However, from 12 to 17 h the temperature of the bare soil was higher and significant (31.2, 32.7 and 32.3 °C, respectively) differing from the temperature of the vegetated soil (27.7, 29.1 and 30.0 °C, respectively). All comparisons between means in the different treatments investigated were supported by Tukey's test at 5% probability [72].

Soil temperature differed between treatments for all times investigated. Thus, the temperature of the bare soil (27.7, 34.5, 39.9, 41.9 and 38.8 °C, respectively) was higher and significant when compared to the vegetated soil (27.1, 29.2, 31.0, 32.2 and 31.4 °C). When analyzing the peak soil temperature that occurred at 3 pm (15 h) in both treatments, the difference between bare soil and vegetated soil reached 9.7 °C.

Even though the bare soil presented a higher soil moisture content (Fig. 3c) in this third moment (DOY 313), the soil temperature differed at all times evaluated (Fig. 3f), being higher and significant in the bare soil (26.9, 31.9, 35.7, 37.8

and 35.5 °C) when compared to vegetated soil (26.3, 28.2, 29.1, 30.1 and 29.2 °C) respectively (Fig. 3f). Similarly to the second moment (DOY 297), the soil temperature peak occurred at 3 pm (15 h). The difference in soil temperature between treatments during the peak was 7.7 °C. Comparisons between means for the third moment were also supported by the Tukey's test at 5% probability [72].

Relationship between Soil Gas Dynamics and Environmental Conditions

This study pointed out some significant weak-moderate to strong linear correlations (Figs. 5 and 6). The inferences about the interpretation of the linear correlation coefficients (r) adopted in this description of the results were adapted from [58, 59]. The so-called bare soil treatment recorded a strong linear correlation between CO₂ emission and soil moisture ($r = 0.76$, $p < 0.05$) (Fig. 5a). While, the vegetated soil treatment, a positive and strong linear correlation was also observed ($r = 0.80$, $p < 0.05$) (Fig. 5d).

Soil CO₂ emission and atmospheric pressure in the investigated treatments. For bare soil, there is a strong and negative correlation between soil CO₂ emission and atmospheric pressure ($r = -0.72$, $p < 0.05$) (Fig. 5b). In the vegetated soil, a moderate correlation was observed between CO₂ emission and atmospheric pressure ($r = -0.60$, $p < 0.05$) (Fig. 5e). However, relative humidity (RH) showed weak-moderate

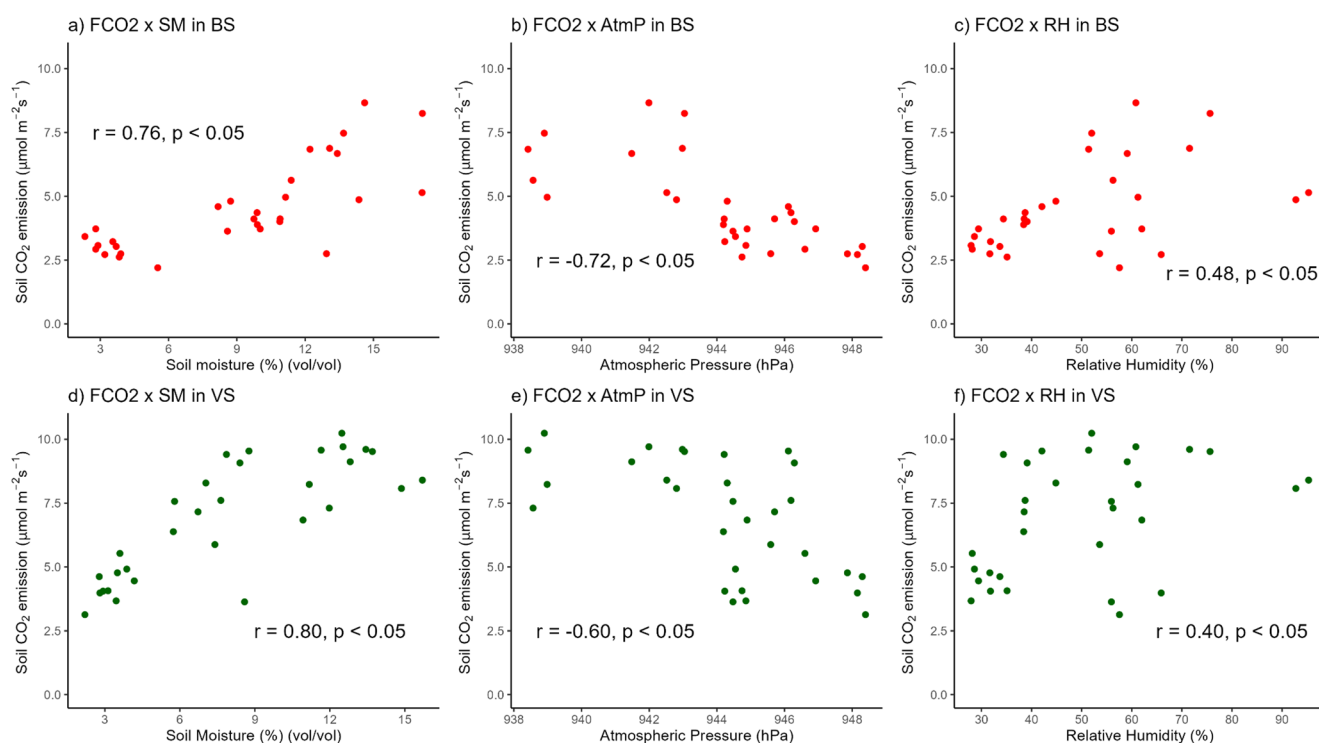


Fig. 5 Pearson's linear correlations between soil CO₂ emission (FCO₂) and environmental variables: (a, d) soil moisture (SM), (b, e) atmospheric pressure (AtmP), and (c, f) relative humidity (RH). Panels

(a–c) represent bare soil (BS), and panels (d–f) represent vegetated soil (VS). Non-significant correlations at the 5% probability level were omitted

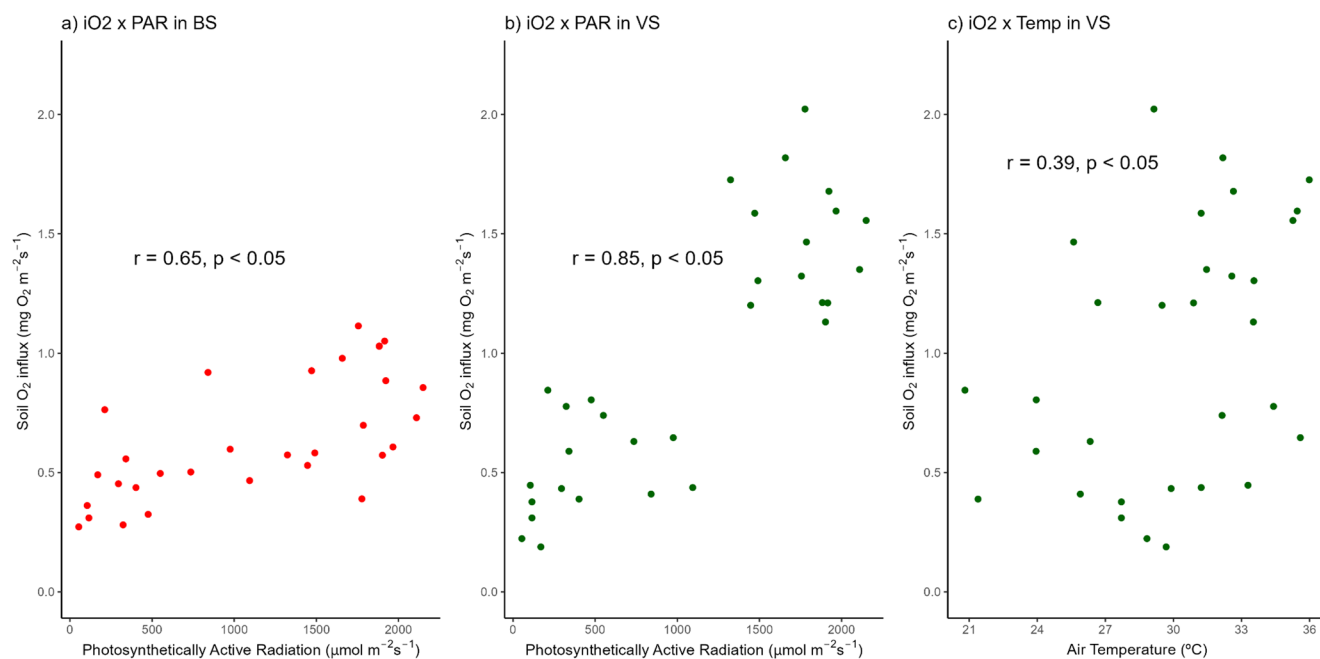


Fig. 6 Pearson's linear correlation for soil O₂ influx and environmental variables. With photosynthetically active radiation (a, b), with air temperature (c). Note: BS is bare soil, VS is vegetated soil. Not significant correlations at 5% probability were omitted

correlations for both treatments, with bare soil ($r=0.48$, $p<0.05$) (Fig. 7c) and vegetated soil ($r=0.40$, $p<0.05$) (Fig. 5f).

Despite the results obtained for CO₂ emission, for the soil O₂ influx the scenario was different, with only one variable being significant in the tests, namely, photosynthetically active radiation for both treatments. While, the relationship between air temperature (Fig. 6c) and soil oxygen influx was significant only in the vegetated soil condition, but it is a weak-moderate relationship ($r=0.38$, $p<0.05$). A moderate positive linear correlation was observed between photosynthetically active radiation and O₂ influx in bare soil ($r=0.65$, $p<0.05$) (Fig. 6a). However, in vegetated soil, a strong and positive correlation was recorded between oxygen influx and photosynthetically active radiation ($r=0.85$, $p<0.05$) (Fig. 6b).

When crossing the correlation data (Figs. 5 and 6) with the hierarchical partitioning analysis (Fig. 4), photosynthetically active radiation was shown to explain 41.24% of the variance that occurs at O₂ influx in bare soil (Fig. 4a). While, in vegetated soil, the explanation of the variance of this phenomenon reaches 51.89% (Fig. 4b). However, the analysis detected a low but significant percentage of air temperature in explaining the O₂ influx in vegetated soil, around 11% (Fig. 4b).

For the CO₂ emission in bare soil, the hierarchical partitioning indicated that the same variables already reported in the correlation were significant, however, in the following

order of importance, atmospheric pressure (30.64%), soil moisture (29.08%) and relative humidity (15%) (Fig. 4c). Despite the ranking, according to the correlation analysis, soil moisture and relative humidity drive CO₂ emission in bare soil, while atmospheric pressure acts inversely (Fig. 5b).

However, the CO₂ emission in vegetated soil, the hierarchical partitioning recorded significant only soil moisture (45.68%) followed by atmospheric pressure (23.13%) (Fig. 7d). While soil moisture drives the soil CO₂ emission, atmospheric pressure acts inversely, in a similar way to what is observed in bare soil, but with a lower percentage of variance explanation in the case of vegetated soil.

Although the hierarchical partitioning indicated slightly subtle numerical divergences between atmospheric pressure (30.64%) and soil moisture (29.08%) in the bare soil condition, soil moisture was considered as a criterion for the importance of the predictor, as it has a higher numerical value in module in the linear correlation analysis ($r=0.76$) when compared to atmospheric pressure ($r = -0.72$) (Fig. 5). Thus, this discussion helped in choosing the best predictors that were used in the following section. For soil CO₂ emission, soil moisture can be used due to the associations reported here, as for soil oxygen influx, the environmental variable selected was photosynthetically active radiation according to the analyzes presented.

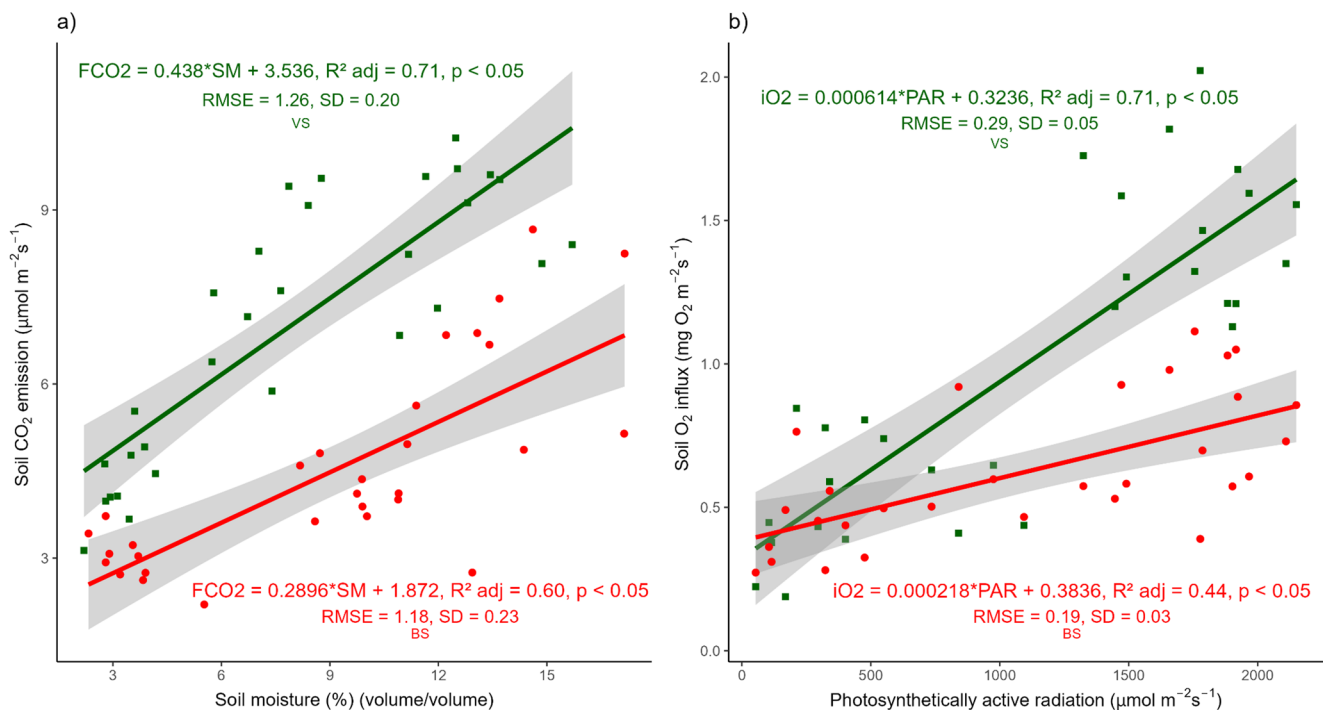


Fig. 7 Linear models with cross validation by K-fold repetition method. In: **a** for soil CO₂ emission as a function of soil moisture (SM), **b** for soil O₂ influx as a function of photosynthetically active radiation (PAR). For bare soil (BS) and vegetated soil (VS). At 5% probability

Table 1 Linear model with cross validation by K-fold repetition method for soil CO₂ emission ($\mu\text{mol m}^{-2} \text{s}^{-1}$) as a function of soil moisture (%) (volume/volume) in vegetated soil

Parameters	Estimated	SEE	RMSE $\pm$ SD	R ²	R ² adj
Intercept	3,536	± 0.488			
SM	0.38	± 0.054	1.26 $\pm$ 0.20	0.72	0.71

Where, SEE is standard error of the estimate, RMSE is root mean squared error followed by the standard deviation (SD) in $\mu\text{mol m}^{-2} \text{s}^{-1}$, R² is coefficient of determination, R² adj is adjusted coefficient of determination. At 5% probability was adopted as level for all parameters

Table 2 Linear model with cross validation by K-fold repetition method for soil CO₂ emission ($\mu\text{mol m}^{-2} \text{s}^{-1}$) as a function of soil moisture (%) (volume/volume) in bare soil

Parameters	Estimated	SEE	RMSE $\pm$ SD	R ²	R ² adj
Intercept	1.872	± 0.473			
SM	0.2896	± 0.046	1.18 $\pm$ 0.23	0.61	0.60

Where, SEE is standard error of the estimate, RMSE is root mean squared error followed by the standard deviation (SD) in $\mu\text{mol m}^{-2} \text{s}^{-1}$, R² is coefficient of determination, R² adj is adjusted coefficient of determination. At 5% probability was adopted as level for all parameters

Table 3 Linear model with cross validation by K-fold repetition method for soil O₂ influx ($\text{mg O}_2 \text{m}^{-2} \text{s}^{-1}$) as a function of photosynthetically active radiation ($\mu\text{mol m}^{-2} \text{s}^{-1}$) in vegetated soil

Parameters	Estimated	SEE	RMSE $\pm$ SD	R ²	R ² adj
Intercept	0.3236	± 0.09937			
PAR	0.000614	± 0.00007521	0.29 $\pm$ 0.05	0.72	0.71

Where, SEE is standard error of the estimate, RMSE is root mean squared error followed by the standard deviation (SD) in $\text{mg O}_2 \text{m}^{-2} \text{s}^{-1}$, R² is coefficient of determination, R² adj is adjusted coefficient of determination. At 5% probability was adopted as level for all parameters

Simple Linear Regression for Soil CO₂ Emission and Soil O₂ Influx

The results in the linear correlation analysis demonstrated a moderate to strong relationship between soil CO₂ emission (FCO₂) and soil moisture (SM), in addition to an association between soil O₂ influx (iO₂) and photosynthetically active radiation (PAR) between the treatments studied, thus the linear models were tested and were significant ($p < 0.05$).

Soil CO₂ emission in vegetated soil showed a linear coefficient of $3.536 \mu\text{mol m}^{-2} \text{s}^{-1}$, unlike bare soil with $1.872 \mu\text{mol m}^{-2} \text{s}^{-1}$. In practical terms, the vegetated soil responded to the increase in soil moisture at a rate of 0.438 (angular coefficient), unlike bare soil which has a rate of only 0.2896; in comparison, this represents a growth rate of vegetated soil that is about 1.5 times higher compared to bare soil (Fig. 7a).

The model also proved to be more accurate in predicting CO₂ emissions in vegetated soil (R² adj=0.71) when compared to bare soil (R² adj=0.60). This indicates an accuracy

Table 4 Linear model with cross validation by K-fold repetition method for soil O₂ influx ($\text{mg O}_2 \text{m}^{-2} \text{s}^{-1}$) as a function of photosynthetically active radiation ($\mu\text{mol m}^{-2} \text{s}^{-1}$) in bare soil

Parameters	Estimated	SEE	RMSE $\pm$ SD	R ²	R ² adj
Intercept	0.3836	± 0.06401			
PAR	0.000218	± 0.00004844	0.19 $\pm$ 0.03	0.46	0.44

Where, SEE is standard error of the estimate, RMSE is root mean squared error followed by the standard deviation (SD) in $\text{mg O}_2 \text{m}^{-2} \text{s}^{-1}$, R² is coefficient of determination, R² adj is adjusted coefficient of determination. At 5% probability was adopted as level for all parameters

of 71% in predicting FCO₂ in vegetated soil versus 60% in bare soil, that is, an 11% greater accuracy. On the other hand, despite there being a numerical difference, there is no evidence that the analyzed models have a different Root Mean Square Error (RMSE), thus, the vegetated soil presented an RMSE of $1.26 \mu\text{mol m}^{-2} \text{s}^{-1}$ (Table 1) against a RMSE of $1.18 \mu\text{mol m}^{-2} \text{s}^{-1}$ on bare soil (Table 2).

Table 3 presents the regression model for iO₂ as a function of PAR in bare soil, notably, the influx of oxygen in bare soil does not seem to be as well explained in relation to vegetated soil (Table 3). While bare soil responds to a rate in relation to the increase in photosynthetically active radiation of 0.000218, in vegetated soil this rate reaches 0.000614. In vegetated soil, PAR explained 71% of the variation in iO₂ (R² adj=0.71), unlike bare soil, in which PAR explained only 44% (R² adj=0.44). In both cases, no differences were observed in the linear coefficients, that is, when the PAR is zero, the O₂ influx in the bare soil and in the vegetated soil are not different from each other (Fig. 7b) see the intercept (Tables 4 and 3).

Although the root mean squared error (RMSE) was presented for each regression model, in our cases when considering the standard deviation of the RMSE it is observed that there is no evidence for the assumption that the errors are different from each other, which returns to adjusted coefficient of determination (R² adj) as a parameter for evaluating the performance and validation of the linear models analyzed in this study. Thus, RMSE metrics signal proportional errors in models when estimating carbon dioxide emission values, but for soil oxygen influx under vegetated and non-vegetated conditions, small differences in RMSE were observed.

Discussion

Diurnal Variability, Root Effect and Environmental Conditions

Temporal patterns were observed depending on the environmental conditions on the evaluated days. The emission of

CO₂ in vegetated soil is higher when compared to bare soil. These differences were more expressive and significant, occurring between 10 and 15 h (Fig. 2a, b and c), in this time interval the highest rates of photosynthetically active radiation were observed, in some cases exceeding 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Table S1).

Soil CO₂ emission in the first and third moments (Fig. 2a and c) were influenced by soil moisture (Fig. 3a and c). Soil moisture is a stimulant of microbial activity and root respiration, leading to increased soil CO₂ emissions [17]. CO₂ emissions in vegetated soil are partly attributed to root respiration, which can account for approximately 50% of soil respiration [49]. However, the contribution of root respiration exhibits high variability, ranging from 10 to 90% of total soil respiration [36].

In this study, the contribution of root respiration to soil CO₂ emission was 27.6% in the first moment (DOY 275), 30.4% in the second moment (DOY 297) and 48.1% in the third moment (DOY 313), while the average value of the root respiration component was around 35.4% of soil respiration (considering mean of all moments investigated). Thus, the amplitude values and the average value of the root contribution to soil respiration indicated in this study corroborate the values reported by [36]; and the theoretical limits of [49].

While, in the second moment (DOY 297), the lower soil moisture (less than 4.0%, Fig. 3b) negatively influenced the CO₂ emission from the soil in both treatments (Fig. 2b). In soil and plants, biochemical and respiratory processes and microbial biomass dynamics are dependent on moisture [20, 81].

Thus, low moisture conditions place stress on soil biochemical processes, and are therefore a limiting factor for the activity of microorganisms in the soil and rhizosphere [59]. In addition, roots depend on water potential for the absorption and transport of water and nutrients, as water potential decreases (lower soil moisture), the difficulty of absorption and energy expenditure increases, which can interrupt photosynthesis. in the aerial part of the plant [20, 42].

Soil O₂ influx were observed between 10 and 15 h (Fig. 2d–f), not by chance, at this time the highest values of photosynthetically active radiation (PAR) also occurred. (Table S1). In plants, the stimulus for root respiration is coupled to increased radiation, which acts as a stimulant for photosynthetic activity [20]. The results suggest that the greater O₂ influx in vegetated soil is influenced by root respiration.

Even at the first moment (DOY 275) the soil O₂ influx did not differ between the treatments for 7 and 17 h. It is possible that this effect is related to the reduction in photosynthetic activity, since photosynthesis is associated with increased levels of photosynthetically active radiation [20].

However, the drop in root respiration at 7 and 17 h may be the effect of the reduction or cessation of photosynthesis and may be attributed to photoinhibition [24]. While, on bare soil, such a drop may also be associated with a lack of solar radiation that conditions a higher soil temperature that facilitates the entry of oxygen into the soil [56].

Photoinhibition occurs when the photon energy required for photosynthesis is at low levels, resulting in a drop in photosynthetic rates that results in low cellular respiration [24, 81, 85]. Precisely at 7am and 5pm (17 h) the PAR rates recorded were lower than 500 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Table S1), which may explain the non-significant results of O₂ influx at the aforementioned times.

This behavior was repeated on days 297 and 313, at 7am, 10am and 5pm (17 h). However, at 12:00 and 15:00 the O₂ influx from the vegetated soil was higher and can be attributed to higher rates of root respiration, as during the peak of photosynthesis that occurs close to noon [20]. Plants can secrete around 40% of photosynthesized products (e.g., soluble sugars) into the rhizosphere, which increases the respiratory activity of the roots and the microbial community present; this process is called the rhizosphere effect [59].

Soil moisture on day 275, in both treatments, began readings in the same range of soil moisture (~ 15.5%), however, throughout the day, the treatment with vegetated soil demonstrated a loss of more soil moisture compared to bare soil, this may be associated with more accelerated evapotranspiration due to photosynthetic activity [65]. Although at the end of the day (at 17 h) both treatments present the same level of humidity (Fig. 3a), this can be explained due to the effect of uncontrolled evaporation that occurs in soils without physical protection (bare soil) to the effects of radiation and temperature [21, 26].

Soil moisture in the second moment (day 297) showed greater fluctuations (Fig. 6b). These variations, in bare soil, can be attributed to the increase in water deficit (low moisture) which is linked to the increase in air temperature and radiation, thus, the water retention capacity in the soil presents high variability [35]. While, in vegetated soil, water potential appears to govern soil moisture conditions throughout the day, which could have resulted in a decrease in evapotranspiration in this treatment and maintenance of soil water potential [29, 64].

Moisture on the third moment (day 313) demonstrated typical behavior after soil moisture recompositing (Fig. 6c). Water loss in vegetated soil comes from the sum of evaporation plus plant transpiration, which results in more water loss throughout the day, while in bare soil water loss occurs only through evaporation [18]. It is worth mentioning that this phenomenon was only observed due to precipitation events that favored the restoration of water content in the soil [25, 83].

The soil temperature increased throughout the day at the different moments evaluated (Fig. 3d–f), thus, the soil temperature showed similar behavior to the air temperature (Table S1) and both temperatures reached their respective values maximum values at 3pm (15 h). Thus, it was observed that the soil temperature responded to the increase in air temperature and radiation (Table S1).

A contrast was observed in soil temperatures between treatments. With higher moisture on day 275, the difference between bare soil and vegetated soil was only 2.8 °C, however, on the second moment (day 297) with lower average soil moisture (approximately 4%), the soil vegetated soil recorded a soil temperature 9.7 °C lower than bare soil. While, in the third moment (day 313) this difference was 7.7 °C. This temperature contrast between bare soil and vegetated soil can be attributed to the effect of root acclimation [19, 47].

The effect of soil temperature under different soil cover conditions has already been observed by [22] who found a decrease in temperature in soils covered by straw or vegetated in relation to bare soils. It is known that different soil covers can offer more or less physical protection to radiation [4, 86], in fact, the soil temperature that responds linearly to the increase in air temperature can be conditioned by the use of vegetation covers and thus alter the influence of radiation on soils [45].

Relationship Between Soil Respiratory Dynamics and Environment

The environmental conditions guided by this discussion are represented by meteorological variables, such as air temperature, relative humidity, atmospheric pressure and photosynthetically active radiation (Table S1). The observed correlations denote that regardless of the treatment, the conditions imposed by the environment tend to govern the soil respiratory dynamics [17, 21].

Soil CO₂ emission and recorded moisture can be explained by the fact that over time the conditions imposed for soil moisture are a conditioning factor for microbial activity [41, 47], however, this effect was already expected and was observed by [1, 52]. Thus, soil moisture is an intrinsic factor associated with soil CO₂ emissions, as it is linked at the same time to microbial biomass activity and plant and root respiration [4, 11, 13], which justifies the strong correlation in vegetated soil ($r = 0.80$; $p < 0.05$). This association of the soil microbiota with the soil CO₂ emission occurs in time and space through the biogenic degradation of soil organic matter and decomposition and this explains the strong correlation also observed in bare soil ($r = 0.76$, $p < 0.05$) [13, 47, 52].

Soil CO₂ emission and atmospheric pressure in the treatments suggest that CO₂ emission decreases as atmospheric pressure increases during the day. However, few studies have proposed relating CO₂ emissions to climate data. In these studies, negative relationships were also observed between CO₂ emission and atmospheric pressure [13, 22, 75]. This behavior may be a reflection of the physical process that describes the flow of gases in the soil, diffusion, described in Fick's first Law in which the transport of air masses suffers from the pressure exerted by the atmosphere, which results in a small drop in emissions of soil gases [13, 56].

This decrease in CO₂ emissions suggested by [13, 56], in our results, demonstrate that the diurnal variability of atmospheric pressure can explain around 30.6% of CO₂ emissions on bare soil and approximately 23.1% of the temporal variability of emissions in vegetated soils as observed in the hierarchical partitioning analysis, it is worth highlighting that these values were significant ($p < 0.05$) (Fig. 4c, d).

Weak or weak-moderate correlations have already been observed in the relationship between relative humidity and soil CO₂ emissions [14]. Thus, environmental meteorological conditions, such as temperature and relative humidity, can affect the process of carbon dioxide emission from the soil to the Earth's atmosphere [14, 21].

Regarding the soil O₂ influx, the effects of atmospheric pressure and temperatures (of air and soil) are already known, as these are factors that facilitate the diffusion of oxygen in soils [22, 56] (Fig. 4c). However, the effect of photosynthetically active radiation in enhancing the influx of oxygen from the atmosphere to the soil has already been reported [56, 75].

It is believed that the demand for oxygen in the soil is increased with cellular respiration and the activation of roots conditioned by photosynthetic activity that depends, among other factors, on water, nutrients and the absorption of sunlight (photon) as a stimulant of this light phase of photosynthesis [20, 59, 65]., this may justify the strong positive correlation between O₂ influx and photosynthetically active radiation in vegetated soil ($r = 0.85$, $p < 0.05$) (Fig. 6b).

In addition, the significant result of the explanation of the variability pointed out by hierarchical partitioning (51.89%) (Fig. 4b) exemplifies the relationship between oxygen influx and photosynthetically active radiation, especially in vegetated soil, which reinforces the possibility of access to root respiration and the stimulation of photosynthesis conditioned by radiation [17, 47, 59].

Soil Respiration Linear Prediction Models

These results advanced the proposition of a simple methodology to predict soil respiration from linear regressions that

use soil moisture and photosynthetically active radiation as inputs, the importance of which has already been mentioned in previous sections. The results of K-fold cross validation with repetition proved to be solid enough for the evaluated database [77, 87].

The main studies point out that the ideal soil temperature for root respiration is between 20 and 30 °C [4, 26, 36] this temperature range is easily observed in tropical conditions [38]. Thus, the modeling of soil CO₂ emission can be carried out using soil moisture as a governing factor of two fundamental compartments in soil respiration, which are microbial activity and the rhizosphere effect [26, 59].

Soil CO₂ emissions of 61% for bare soil and 71% for vegetated soils. The statistical inferences were also strong enough not to reject the hypothesis of differences between the growth rate (angular coefficient) when comparing bare soil versus vegetated soil (Tables 1 and 2). Previously [67], had proposed prediction using multiple linear regression based on soil attributes, including soil moisture, and [14] using several soil and climate variables through machine learning. Thus, the estimates predicted in this linear modeling appeared promising, simple and with potential for greater application in promoting greater sustainability in agricultural systems, forestry plantations and cultivated plants [22, 68].

This study presents an unprecedented result in the literature, an estimate of soil oxygen influx through a simple linear regression [75]. had proposed an estimate using artificial neural networks, but, according to the authors, the database was not sufficient to obtain a general model. In this way, this study contributes to the understanding of the dynamics of global soil respiration. In bare soil, without the action of the rhizosphere effect [53, 59], only a 44% explanation of O₂ influx was obtained, but when analyzing vegetated soil, the explanation is 71% (Fig. 7b).

The range in estimating soil oxygen comprises all ranges already reported in studies by [1] in sugar cane; [74] in planted forests and by [22] in different soil covers. This is important information that suggests that this model is possible to predict the soil O₂ influx under different conditions. The models presented in this study showed greater capacity to predict the O₂ influx when compared to that reported by [75], despite the use of neural networks.

Estimating CO₂ emissions and O₂ influx into soils, especially tropical ones, is an important indicator that helps in the adoption of more sustainable agricultural practices [22, 51, 74], in addition to allowing the creation of a new indicator, they can collaborate in the creation of a new indicator, called soil respiratory quotient, that is, the ratio between CO₂ efflux and O₂ influx into the system, which helps in understanding the dynamics of global soil respiration [1, 3, 82].

Conclusions

The results indicate that root respiration is correlated with soil O₂ influx, with root respiration representing 27 to 48% of soil respiration.

Soil CO₂ emissions respond to atmospheric pressure in both vegetated and bare soils, with atmospheric pressure explaining 20 to 30% of the diurnal variability in CO₂ emissions.

Soil CO₂ emissions, with and without vegetation, were predicted using a linear model based on soil moisture.

The O₂ influx vegetated soil can be predicted as a function of photosynthetically active radiation (PAR).

In this way, this study contributed to the progress of understanding soil respiration and the contribution of root respiration, by proposing a simplified approach to predicting two important gases that regulate global soil respiration. However, the limitation of this measurement method is the time spent on each reading (~5 min per point), which makes it difficult to perform over large areas. However, a validation strategy for the proposed models for large areas can be subsequently used through remote sensing.

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Data Availability The data that support the findings of this study are available from the corresponding author (De Lucena, W.B.) upon reasonable request.

Declarations

Conflict of Interest The authors declare no competing interests.

Ethical Approval All authors have read, understood, and have complied as applicable with the statement on “Ethical responsibilities of Authors” as found in the Instructions for Authors and are aware that with minor exceptions, no changes can be made to authorship once the paper is submitted.

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